02 / ALGEBRA / EX25

Engineering & mathematical theory

Vectors, components, norms, and their connection to force addition.

DOCUMENT 01 / THEORY

What you will learn

- Define vectors in two and three dimensions and distinguish them from scalars.
- Develop vector addition, subtraction, scalar multiplication, and norms.
- Work through pure mathematical examples before introducing physical forces.

Prerequisites: Variables, basic arithmetic, lists, and Cartesian axes.

Reading order: 1. Theory → 2. Reference → 3. Exercise → 4. Pseudocode → 5. Programming.

1. What are we studying?

EX25 begins with **vector algebra**. We first define operations without physical units, then use exactly the same mathematics to describe forces in a plane. The algebra does not depend on whether the vector describes force, velocity, or position.

2. Scalars, vectors, and dimension

A scalar is a single number. A vector can be represented by ordered components once a basis is chosen:

$$\mathbf{v} = (v_x, v_y) \in \mathbb{R}^2$$

A three-dimensional vector has three components:

$$\mathbf{v} = (v_x, v_y, v_z) \in \mathbb{R}^3$$

Bold letters denote vectors. Their individual components are scalars. NumPy arrays will be our initial representation in code.

3. Cartesian basis and unit vectors

Unit basis vectors have length one and point along the coordinate axes:

$$\hat{\mathbf{i}} = (1, 0), \qquad \hat{\mathbf{j}} = (0, 1)$$

$$\mathbf{v} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} \quad \Longleftrightarrow \quad \mathbf{v} = (v_x, v_y)$$

> The symbols **î** and **ĵ** denote Cartesian **unit vectors**, not the imaginary unit used in complex numbers. The same vector can be written as a column of components.

4. Vector operations

Addition and subtraction act component by component and require equal dimensions:

$$\mathbf{u} + \mathbf{v} = (u_x + v_x,\ u_y + v_y)$$

$$\mathbf{u} - \mathbf{v} = (u_x - v_x,\ u_y - v_y)$$

Multiplication by a scalar changes each component:

$$\alpha\mathbf{v} = (\alpha v_x, \alpha v_y)$$

A linear combination is central to linear algebra:

$$\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v}$$

Important laws include commutativity, associativity, the zero vector, additive inverses, and distributivity:

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$\alpha(\mathbf{u}+\mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$$

5. What is a norm?

A norm assigns each vector a non-negative real magnitude. Its defining properties are positivity with zero only at the zero vector, homogeneity, and the triangle inequality.

The Euclidean or **2-norm** measures length. In two dimensions, it follows from Pythagoras:

$$\|\mathbf{v}\|_2 = \sqrt{v_x^2 + v_y^2}$$

In n dimensions:

$$\|\mathbf{v}\|_2 = \sqrt{\sum_{i=1}^{n} v_i^2}$$

Two other commonly used vector norms are:

$$\|\mathbf{v}\|_1 = \sum_{i=1}^{n} |v_i|$$

$$\|\mathbf{v}\|_\infty = \max_{1 \le i \le n} |v_i|$$

For (3,4), the 1-norm is 7, the 2-norm is 5, and the infinity norm is 4. Each measures a different aspect of size.

6. Properties of the 2-norm

Homogeneity means scaling a vector scales its magnitude by the absolute value of the factor:

$$\|\alpha\mathbf{v}\|_2 = |\alpha|\,\|\mathbf{v}\|_2$$

The triangle inequality bounds the magnitude of a sum:

$$\|\mathbf{u}+\mathbf{v}\|_2 \le \|\mathbf{u}\|_2 + \|\mathbf{v}\|_2$$

Geometrically, a straight segment cannot be longer than a two-segment path connecting the same endpoints.

7. Fully worked mathematical example

Given: two dimensionless vectors.

$$\mathbf{u} = (3, -1), \qquad \mathbf{v} = (-2, 4)$$

Step 1. Add the components:

$$\mathbf{u}+\mathbf{v} = (3 + (-2),\ -1 + 4)$$

$$\mathbf{u}+\mathbf{v} = (1, 3)$$

Step 2. Compute a linear combination:

$$2\mathbf{u} = (6,\ -2)$$

$$2\mathbf{u} - \mathbf{v} = (6,\ -2) - (-2, 4)$$

$$2\mathbf{u} - \mathbf{v} = (8,\ -6)$$

Step 3. Compute the 2-norms:

$$\|\mathbf{u} + \mathbf{v}\|_2 = \sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.1623$$

$$\|2\mathbf{u} - \mathbf{v}\|_2 = \sqrt{8^2 + (-6)^2} = 10$$

Step 4. Check the triangle inequality:

$$\|\mathbf{u}\|_2 = \sqrt{10}, \qquad \|\mathbf{v}\|_2 = \sqrt{20}$$

$$\sqrt{10} \le \sqrt{10} + \sqrt{20}$$

This example is fully solved **without units or forces**. The same steps apply to the force vectors below.

8. Physical application: two forces in a plane

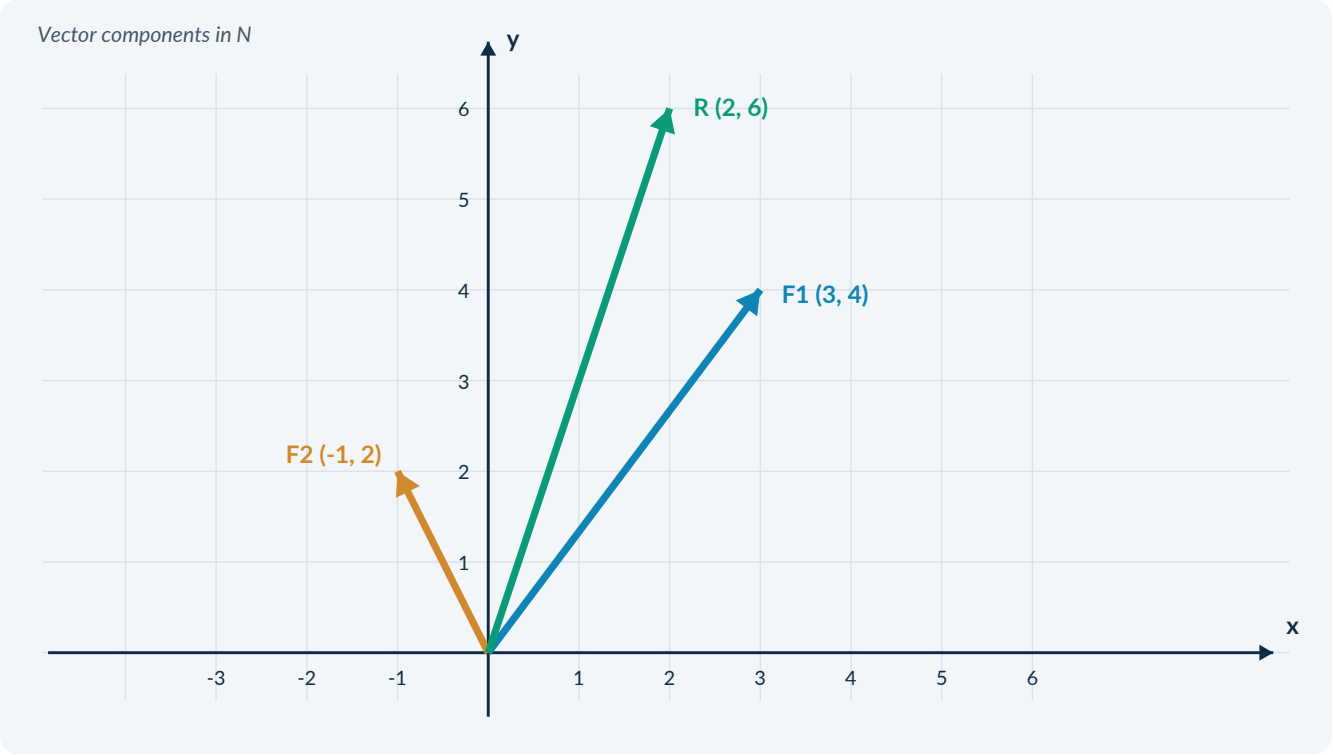
Force is a vector: it has components along coordinate axes. Consider two forces expressed in a Cartesian basis:

$$\mathbf{F}_1 = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}})\ \text{N}$$

$$\mathbf{F}_2 = (-1\hat{\mathbf{i}} + 2\hat{\mathbf{j}})\ \text{N}$$

Basis vectors are dimensionless; their coefficients are force components in newtons. Component notation is equivalent:

$$\mathbf{F}_1 = (3, 4)\ \text{N}, \qquad \mathbf{F}_2 = (-1, 2)\ \text{N}$$



Resultant: add the horizontal and vertical components separately.

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 = (3 - 1,\ 4 + 2)\,\mathrm{N}$$

$$\mathbf{R} = (2, 6)\,\mathrm{N}$$

$$\|\mathbf{R}\|_2 = \sqrt{2^2 + 6^2} = \sqrt{40} \approx 6.3249\,\mathrm{N}$$

The resultant magnitude is no greater than the sum of the individual force magnitudes. This is a useful physical and algebraic check.

9. Connecting the theory to Python

Use **np.array** to represent vectors, + and - for componentwise operations, multiplication for scaling, and **np.linalg.norm** for magnitudes. The reference guide explains syntax; the exercise asks you to choose and connect these tools.

10. Key points

- **Vector:** components relative to a basis, written in bold.
- **2-norm:** Euclidean length, a non-negative scalar.
- **Force:** a physical use of the same algebra, with units of N.
- **Validation:** manual calculations, component signs, and plausible magnitudes.